

- ⑪ I varies directly with E and inversely with R .
If

$I = 3$ when $E = 240$ and $R = 40$, find the equation that connects I , E and R . Hence find I when $E = 120$ and $R = 30$.

- ⑫ If $\mathbb{E} = \{\text{positive integers}\}$, list the members of the following sets.

- a $\{x : x > 9\}$
- b $\{x : x > -9\}$
- c $\{x : 2x = 11\}$

Geometry and mensuration

- a Fig. P7 gives sketches and names of some common solids.

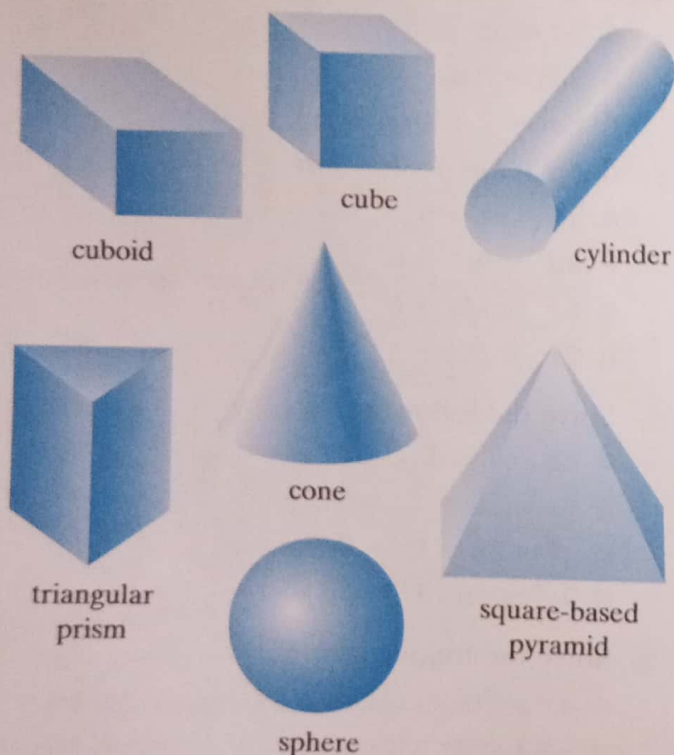


Fig. P7

All solids have **faces**; most solids have **edges** and **vertices** (Fig. P8).

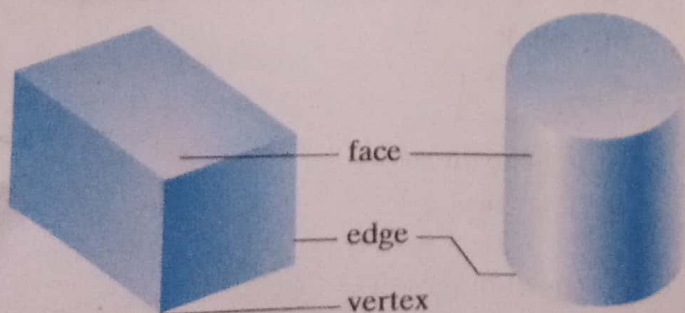


Fig. P8

Formulae for the **surface area** and **volume** of common solids are given in Table P1.

	surface area	volume
cube edge s	$6s^2$	s^3
cuboid length l , breadth b , height h	$2(lb + bh + lh)$	lbh
prism height h , base area A		Ah
sphere radius r	$4\pi r^2$	$\frac{4}{3}\pi r^3$
cylinder base radius r , height h	$2\pi r(r + h)$	$\pi r^2 h$
cone base radius r , height h , slant height l	$\pi r(r + l)$	$\frac{1}{3}\pi r^2 h$

Table P1

Fig. P9 shows two common methods of **drawing** a cuboid. These methods may be used with any solid.

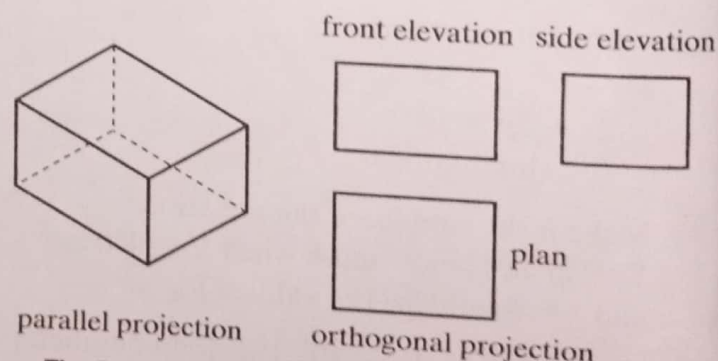


Fig. P9

In **parallel projections**, lines that are parallel on the solid appear parallel on the drawing. In **orthogonal projections**, the solid is represented by separate scale drawings of its **plan** and **elevations**.

The **net** of a solid is the plane shape which can be folded to make the solid. The net of a cube is shown in Fig. P10.

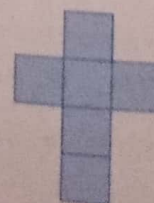


Fig. P10

b Angle is a measure of rotation or turning.

1 **revolution** = 360 **degrees** (1 rev = 360°)

1 **degree** = 60 **minutes** ($1^\circ = 60'$)

The names of angles change with their size.
See Fig. P11.

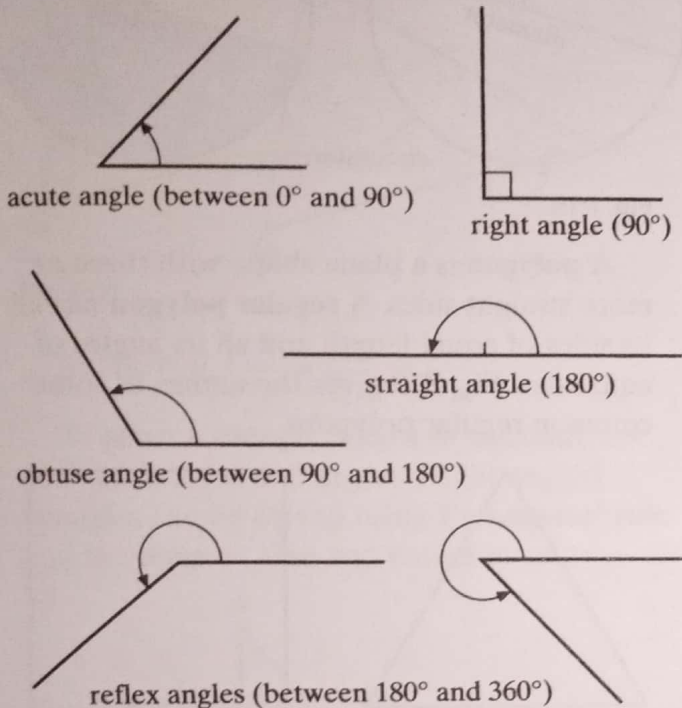


Fig. P11

Angles are measured and constructed using a **protractor**.

Fig. P12 shows some properties of angles formed when straight lines meet.

e) Corresponding angles are equal. $a = b$ and $p = q$

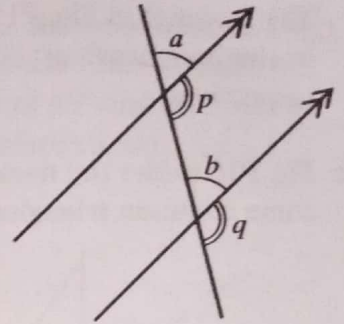


Fig. P12

In Fig. P13, α is the **angle of elevation** of the top of the flag-pole from the girl and β is the **angle of depression** of the girl from the boy.

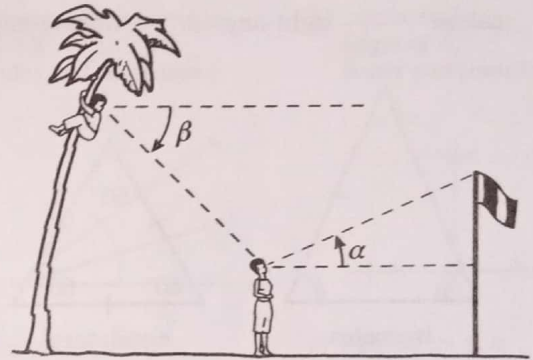


Fig. P13

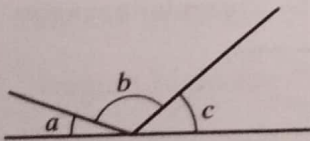
Directions are taken from the **points of the compass** (Fig. P14).



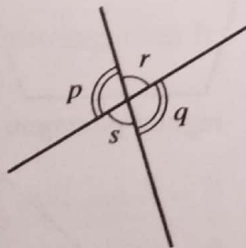
Fig. P14

A **3-figure bearing** is a direction given as the number of degrees from north measured in a clockwise direction. See Fig. P15.

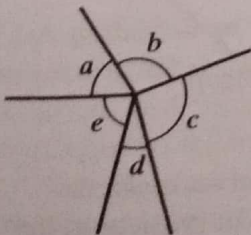
a) The sum of the angles on a straight line is 180° .
 $a + b + c = 180^\circ$



b) Vertically opposite angles are equal.
 $p = q$ and $r = s$



c) The sum of the angles at a point is 360° .
 $a + b + c + d + e = 360^\circ$



d) Alternate angles are equal. $x = y$ and $m = n$

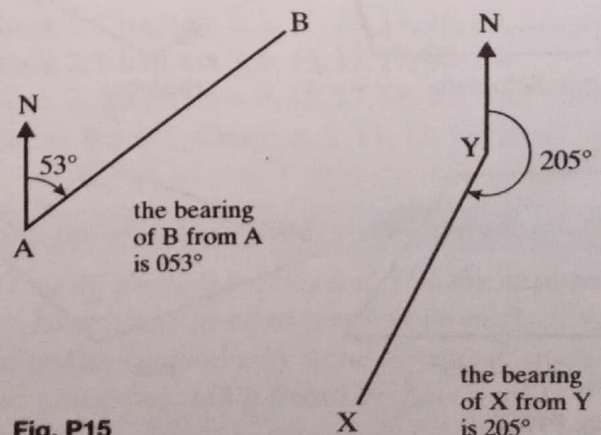
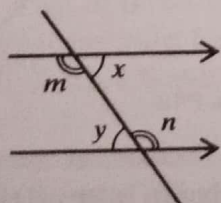


Fig. P15

The bearings in Fig. P15 can also be given as **compass bearings**: N53°E and S25°W respectively.

- c Fig. P16 shows the names and properties of some common **triangles**.

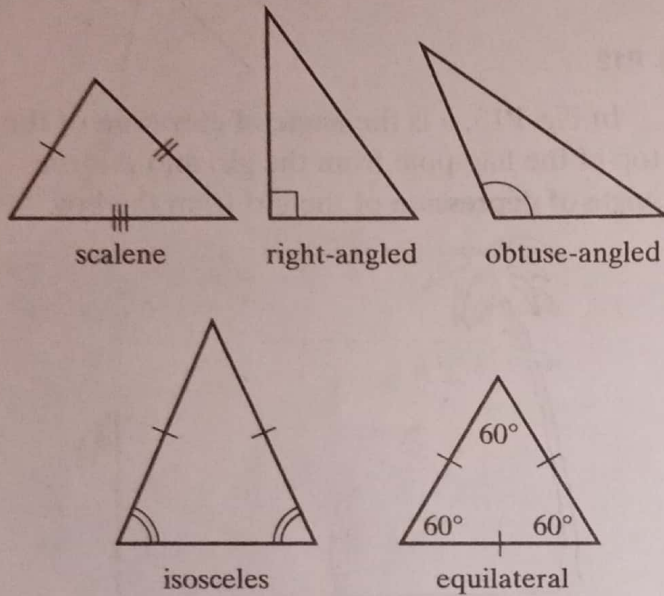


Fig. P16 Some common triangles

Fig. P17 shows the names and properties of some common **quadrilaterals**.

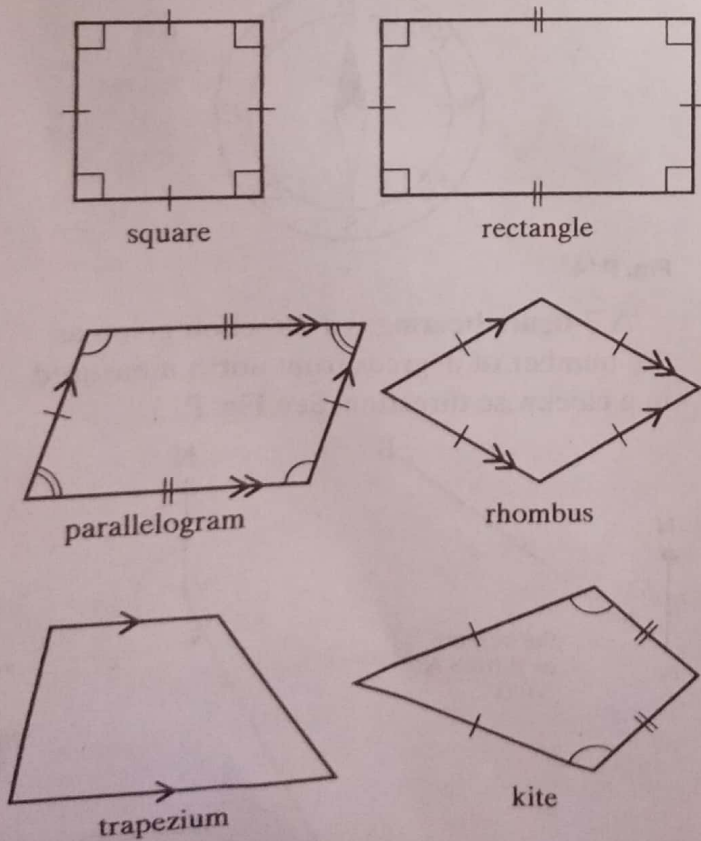


Fig. P17

Fig. P18 gives the names of lines and regions in a **circle**.

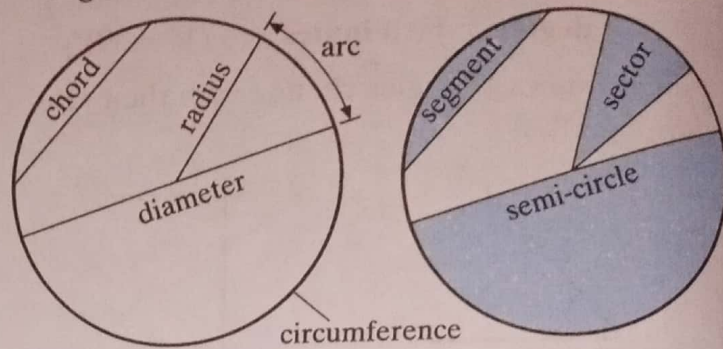


Fig. P18

A **polygon** is a plane shape with three or more straight sides. A **regular polygon** has all its sides of equal length and all its angles of equal size. Fig. P19 gives the names of some common regular polygons.

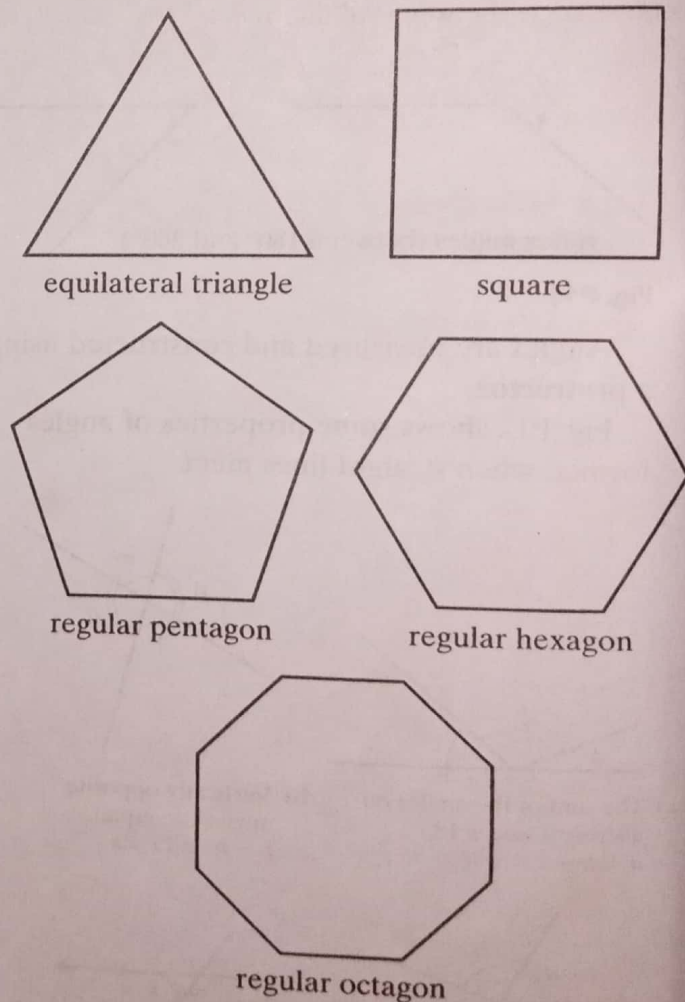


Fig. P19

The **sum of the angles of an n -sided polygon** is $(n - 2) \times 180^\circ$. In particular, the sum of the angles of a triangle is 180° and the sum of the angles of a quadrilateral is 360° .

Any two shapes are **similar** if they are **equiangular** and if their corresponding sides are in the same ratio. Any two triangles are similar if they are equiangular (Fig. P20).

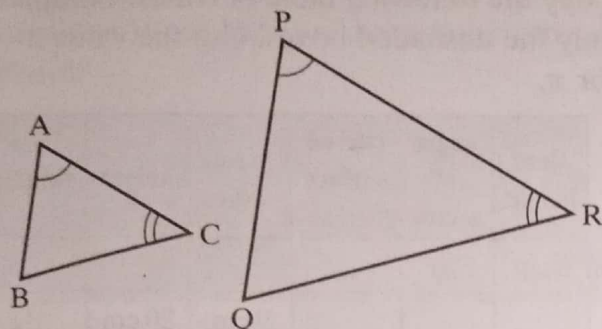


Fig. P20

In Fig. P20, $\frac{AB}{PQ} = \frac{BC}{QR} = \frac{AC}{PR}$

To **solve a triangle** means to calculate the sizes of its sides and angles. Right-angled triangles can be solved using **Pythagoras' rule** and the **tangent, sine** and **cosine** ratios.

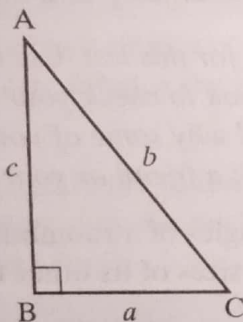


Fig. P21

In Fig. P21, $\triangle ABC$ is right-angled at B; side $\overline{AC}$ is the **hypotenuse**.

Pythagoras' rule: $b^2 = a^2 + c^2$

tangent ratio: $\tan \widehat{C} = \frac{c}{a}$ and $\tan \widehat{A} = \frac{a}{c}$

sine ratio: $\sin \widehat{C} = \frac{c}{b}$ and $\sin \widehat{A} = \frac{a}{b}$

cosine ratio: $\cos \widehat{C} = \frac{a}{b}$ and $\cos \widehat{A} = \frac{c}{b}$

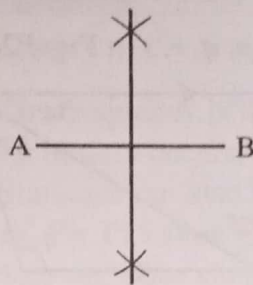
A **pythagorean triple** is a set of three numbers that satisfies Pythagoras' rule. For example {3, 4, 5}, {5, 12, 13}, {8, 15, 17} are pythagorean triples.

Formulae for the **perimeter** and **area of plane shapes** are given in the table on page 233.

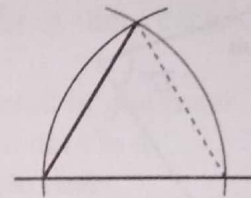
The **SI system of units** is given in the tables on pages 232 and 233.

d The sketches in Fig. P22 show the main features of the common **geometrical constructions**.

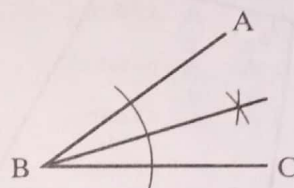
To construct angles of 45° and 30° , bisect angles of 90° and 60° respectively.



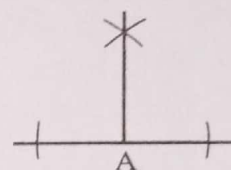
a) perpendicular bisector of AB (ruler and compasses)



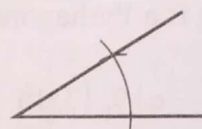
b) equilateral $\triangle$ angle of 60° (ruler and compasses)



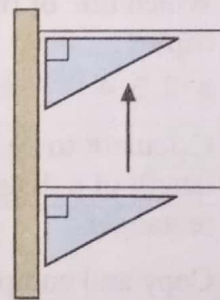
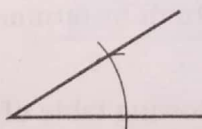
c) bisector of $\widehat{ABC}$ (ruler and compasses)



d) right angle at A (ruler and compasses)



e) to copy an angle (ruler and compasses)



f) parallel lines (ruler and set square)

Fig. P22

Detailed coverage of *geometry and mensuration* is given in the JSS course books as follows:

Book 1, Chapters 6, 8, 11, 13, 14, 16, 20, 21;

Book 2, Chapters 3, 6, 16, 17, 19, 20, 24;

Book 3, Chapters 4, 8, 15, 17, 18, R3, and the SSS course Book 1, Chapters 2, 11, 12, 15, 16, 17.

Review test 5 (Geometry and mensuration)

Allow 40 minutes for this test. Use the answers at the back of the book to check your work. If you do not understand why some of your answers were incorrect, ask a friend or your teacher. Then try Test 6.

- ① One of the angles of a parallelogram is 41° . What are the sizes of its other three angles?
- ② How many lines of bilateral symmetry does a rectangle have?
- ③ Find the values of m, n, p, q, r, s in Fig. P28.

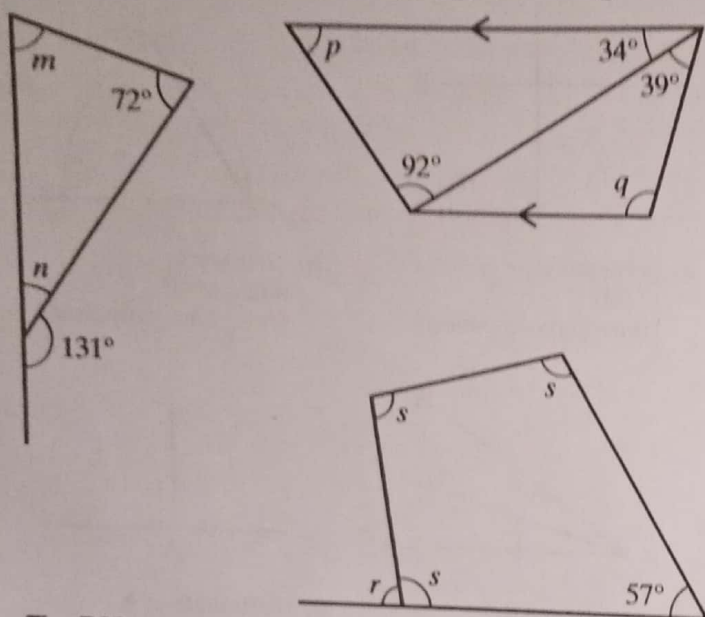


Fig. P28

- ④ Which one of the following is a Pythagorean triple?
a 2, 3, 4 **b** 5, 11, 12 **c** 9, 12, 15
- ⑤ Calculate to 2 significant figures the length of a diagonal of a 30 mm by 60 mm rectangle.
- ⑥ Copy and complete the following table of cylinders. Use the value 3 for π .

	radius	height	curved surface area	volume
a	5 cm	10 cm		
b		18 m	486 m^2	
c	3.5 cm			168 cm^3

Table P3

- ⑦ Use the tables on pages 238–240 to find the value of α in each of the following. Give all answers to the nearest 0.1 of a degree.
a $\tan \alpha = 1.552$ **b** $\tan \alpha = -1.552$
c $\sin \alpha = 0.555$ **d** $\sin \alpha = -0.555$
e $\cos \alpha = 0.33$ **f** $\cos \alpha = -0.33$

- ⑧ $\triangle ABC$ is similar to $\triangle XYZ$. If $AC = 8 \text{ cm}$, $BC = 6 \text{ cm}$ and $YZ = 11 \text{ cm}$, find the length of XZ .
- ⑨ Copy the following table of cones. Complete only the unshaded boxes. Use the value 3 for π .

	slant height	angle of sector	curved surface area	base radius	height	volume
a	8 cm	270°				
b				9 cm	20 cm	

Table P4

- ⑩ On a map of scale 1 : 25 000, a region of a country has an area of 13.5 cm^2 . Find the actual area of the region to the nearest hectare.

Review test 6 (Geometry and mensuration)

Allow 40 minutes for this test. Use the answers at the back of the book to check your work. If you do not understand why some of your answers were incorrect, ask a friend or your teacher.

- ① One of the angles of a rhombus is 102° . What are the sizes of its other three angles?
- ② How many planes of bilateral symmetry does a cube have?
- ③ Find the values of p, q, r, s, t, x in Fig. P29.

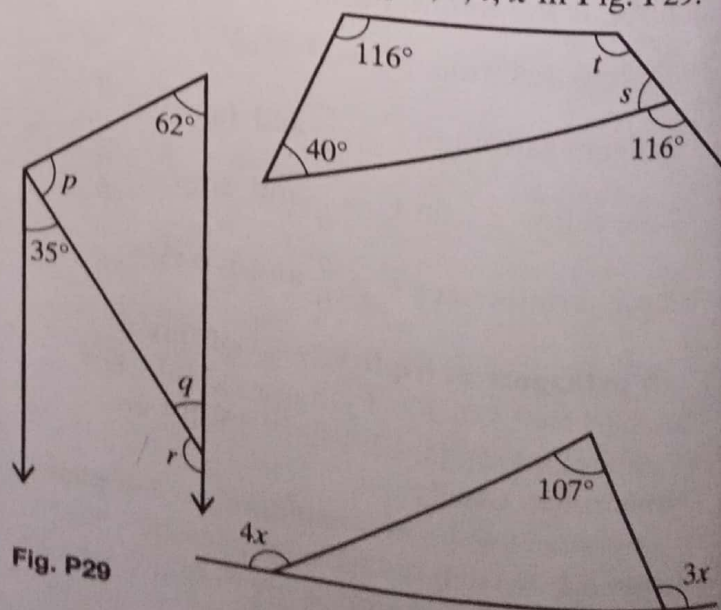


Fig. P29

- ④ Two of the sides of a right-angled triangle are 5 m and 12 m. Find the length of the third side. (There are two possible answers.)

- ⑤ A chord is 6 cm from the centre of a circle of radius 10 cm. What is the length of the chord?

- ⑥ Copy and complete the following table of cylinders. Use the value 3 for π .

	radius	height	curved surface area	volume
a	8 cm	15 cm		
b		14 m	504 m ²	
c	4.5 cm			720 cm ³

Table P5

- ⑦ Use the table on page 219 to find

a $\cos 36^\circ$ b $\cos 136^\circ$

c $\cos 236^\circ$ d $\cos 336^\circ$

- ⑧ The volumes of two similar cones are in the ratio 64:216. What is the ratio of their heights?

- ⑨ Copy and complete the following table of cones. Complete only the unshaded boxes. Use the value 3 for π .

	slant height	angle of sector	curved surface area	base radius	height	volume
a	6 cm	210°				
b				4.5 cm	32 cm	

Table P6

- ⑩ On a map a lake has an area of 3.9 cm². If the actual area of the lake is 15.6 hectares, find the scale of the map.

Statistics and probability

- a Information in numerical form is called **statistics**. Statistical **data** may be given in **rank order** (i.e. in order of increasing size) such as in the following marks obtained in a test out of 5:

0, 1, 1, 2, 2, 2, 3, 3, 5

Data may also be given in a **frequency table** (Table P2).

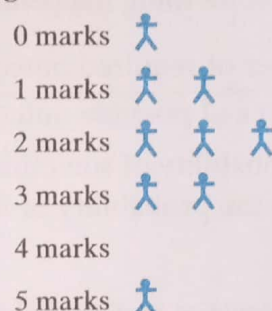
mark	0	1	2	3	4	5
frequency	1	2	3	2	0	1

Table P2

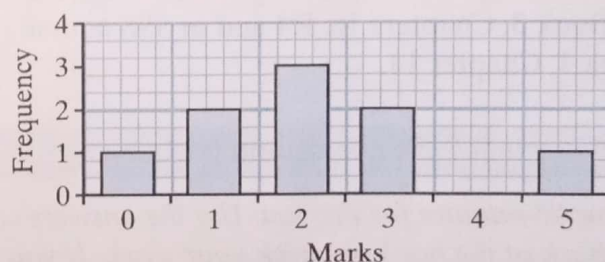
The **frequency** is the number of times each piece of data occurs.

Statistics can also be presented in graphical form. Fig. P25 shows the above data in a **pictogram**, a **bar chart** and a **pie chart**.

pictogram



bar chart



pie chart

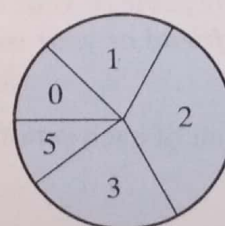


Fig. P25

- b The **average** of a set of statistical data is a number which is representative of the whole set. The three most common averages are the **mean**, the **median** and the **mode**. For the nine numbers given in rank order in paragraph a above,

$$\begin{aligned} \text{mean} &= \frac{0 + 1 + 1 + 2 + 2 + 2 + 3 + 3 + 5}{9} \\ &= 2\frac{1}{9} \end{aligned}$$

The *median* is the middle number when the data are arranged in order of size (2).

Note: If there is an even number of data, the median is the mean of the middle two numbers.

The *mode* is the number with the greatest frequency (also 2 in this case). If a set of data has two modes we say it is **bimodal**.

- c Probability** is a measure of the likelihood of a required outcome happening.

If an event *must* happen, probability = 1

If an event *cannot* happen, probability = 0

If the probability of something happening is p , then

$$\text{probability} = \frac{\text{number of required outcomes}}{\text{number of possible outcomes}}$$

$0 \leq p \leq 1$. If the probability of something happening is x , then the probability of it *not* happening is $1 - x$.

Detailed coverage of *statistics and probability* is given in the JSS course books as follows:

Book 1, Chapters 17, 18, 22; Book 2, Chapters 18, 21, Book 3, Chapters 16, R4 and in SSS course Book 1, Chapter 14.

Review test 7 (Statistics and probability)

Allow 25 minutes for this test. Use the answers at the back of the book to check your work. If you do not understand why some of your answers were incorrect, ask a friend or your teacher. Then try Test 8.

- ① Calculate the mean of each set of numbers.
 - a 14, 16, 3
 - b 11, 12, 16
 - c 0, 6, 6, 6, 7
 - d 8, 8, 4, 3, 12
- ② Find the mode, median and mean of each set of numbers.
 - a 7, 7, 9, 12, 15
 - b 11, 12, 15, 15, 16, 16, 16, 17, 17, 18
 - c 9, 7, 4, 11, 7, 10
 - d 2, 0, 6, 2, 7, 3, 1, 4, 2

- ③ Table P10 shows the results of a survey of vehicles on a city road in one hour.

vehicle	cars	lorries	buses	others	total
frequency	26	9	9	6	

Table P10

- a How many vehicles altogether?
 - b What is the probability that the next vehicle is a car?
 - c What is the probability that the next vehicle is *not* a car?
 - d What is the probability that the next vehicle is neither a car, a lorry nor a bus?
- ④ A number is chosen at random from the set of numbers
 $\{21, 22, 23, \dots, 34, 35\}$.

What is the probability that it is

- a a multiple of 4
- b a prime number
- c the number 18
- d a whole number?

Review test 8 (Statistics and probability)

Allow 25 minutes for this test. Use the answers at the back of the book to check your work. If you do not understand why some of your answers were incorrect, ask a friend or your teacher.

- ① Calculate the mean of each set of numbers.
 - a 24, 12, 14, 18
 - b 0, 7, 4, 5
 - c 4, 10, 0, 5, 8, 0
 - d 10, 11, 12, 13, 15, 17
- ② Find the mode, median and mean of each set of numbers.
 - a 13, 14, 14, 16, 17, 19
 - b 4, 8, 11, 11, 12, 12, 12
 - c 1, 0, 15, 0, 8, 12
 - d 9, 8, 6, 9, 6, 5, 7, 9, 7, 8, 9, 7
- ③ A school enters candidates for the SSCE examination. Table P11 shows the results for the years 2007–2010.

	2007	2008	2009	2010
candidates entered	84	93	111	112
number who passed	54	55	67	72

Table P11

During the four years:

- a** how many candidates were entered for the SSCE?
- b** how many candidates passed?
- c** what is the school's success rate (as a percentage)?
- d** what is the approximate probability (as a decimal to 1 s.f.) of a student at this school passing the SSCE?

- ④ A bag contains 28 red balls and 8 blue balls. A ball is chosen at random. What is the probability that it is:

- a** red
- b** blue
- c** either red or blue
- d** neither red nor blue?

Challenge Box

Arrange these digits:

1, 1, 2, 2, 3, 3, 4, 4

... so that the 1's are four digits apart,
the 2's are three digits apart,
the 3's are two digits apart,
and the 4's are one digit apart.

1

Numerical processes (1) Logarithms

OBJECTIVES

By the end of this chapter you should be able to:

- 1 Recall the use of logarithm tables to perform calculations with numbers greater than 1
- 2 Compare characteristics of logarithms with corresponding numbers in standard form
- 3 Use logarithm tables to perform calculations with numbers less than 1, including
 - multiplication and division
 - powers and roots of numbers
- 4 Solve simple logarithmic equations.

Teaching and learning materials

Teacher: Index and logarithm charts; books of 4-figure tables (as used in public examinations).

Student: Logarithm and antilogarithm tables (provided on pages 236 and 237 of this book).

1.1 Standard form

Table 1.1 shows that fractions such as one tenth, one hundredth, ... can be expressed as negative powers of 10.

number	power of 10
1000	10^3
100	10^2
10	10^1
1	10^0
0.1	10^{-1}
0.01	10^{-2}
0.001	10^{-3}

Table 1.1

A number in the form $A \times 10^n$, where A is a number between 1 and 10 and n is an integer, is said to be in **standard form**. For example, 3.753×10^2 and 8.5×10^{-3} are numbers in standard form.

Example 1

Express the following in standard form.

- a 234 b 4 231 c 0.23 d 0.031

$$\begin{aligned} a \quad 234 &= 2.34 \times 100 \\ &= 2.34 \times 10^2 \end{aligned}$$

$$\begin{aligned} b \quad 4\,231 &= 4.231 \times 1\,000 \\ &= 4.231 \times 10^3 \end{aligned}$$

$$\begin{aligned} c \quad 0.23 &= 2.3 \times 0.1 \\ &= 2.3 \times 10^{-1} \end{aligned}$$

$$\begin{aligned} d \quad 0.031 &= 3.1 \times 0.01 \\ &= 3.1 \times 10^{-2} \end{aligned}$$

Exercise 1a (Oral revision)

Express the following in standard form.

- 1 32.4
- 2 0.471
- 3 3 472 000
- 4 0.000 613 1
- 5 4 576
- 6 51 720
- 7 0.043 81
- 8 0.000 000 231
- 9 623 000 000
- 10 0.003 471
- 11 8.04
- 12 510
- 13 6 500
- 14 24.12
- 15 0.027 9
- 16 0.000 193
- 17 0.102
- 18 12 800
- 19 0.007 43
- 20 500.8

1.2 Logarithms of numbers greater than 1

In Book 1 it was shown that the base 10 logarithm of a number is the power to which 10 is raised to give that number. Tables such

as those on page 236 are used to find the logarithms of numbers. For example,

$$\begin{aligned} 374.5 &= 3.745 \times 10^2 \text{ (standard form)} \\ &= 10^{0.5735} \times 10^2 \text{ (using tables)} \\ &= 10^{0.5735 + 2} \\ &= 10^{2.5735} \end{aligned}$$

$\log_{10} 374.5 = 2.5735$ (said as 'the logarithm to base 10 of 374.5 is 2.5735')

When using logarithms to perform calculations, we assume that the base is 10 and simply write $\log_{10} 374.5$ as $\log 374.5$. See Section 1.6 for logarithms in other bases.

The logarithm of a number is in two parts, the **characteristic** and the **mantissa**. See Fig. 1.1.

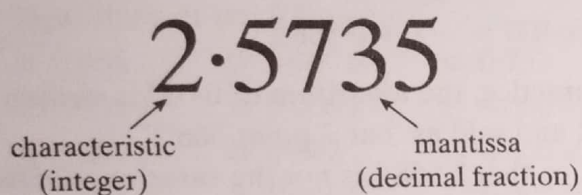


Fig. 1.1

If a number is in standard form, it is easy to see what the characteristic of its logarithm is. For example, $\log 734$ has the characteristic 2 because

$$734 = 7.34 \times 10^2$$

and the characteristic of $\log 5203$ is 3 because

$$5203 = 5.203 \times 10^3.$$

Exercise 1b (Revision)

- a** (Oral) State the characteristics of the logarithms of the following numbers.
- b** Use tables to find the complete logarithms of the following numbers.

- | | |
|-------------|----------|
| ① 900 | ② 90 |
| ③ 9 | ④ 4 620 |
| ⑤ 46.2 | ⑥ 46 200 |
| ⑦ 800.3 | ⑧ 8.003 |
| ⑨ 800 300 | ⑩ 12.34 |
| ⑪ 1 234 000 | ⑫ 1 234 |

Multiplication and division (revision)

To multiply and divide numbers, first express them as logarithms and then apply the addition

and subtraction laws of indices to the logarithms. Set the work out in tables as in Example 2. Use the logarithm tables on page 236 to change the numbers to logarithms. Finally, use the anti-logarithm tables on page 237 to change the logarithms back to numbers.

Example 2

Calculate $\frac{76.7 \times 308.2}{8.04}$.

number		log
76.7	→	1.8848
308.2	→	+2.4889
		4.3737
8.04	→	-0.9053
2941	←	3.4684

$$\frac{76.7 \times 308.2}{8.04} = 2941 = 2940 \text{ to 3 s.f.}$$

Check: $\frac{76.7 \times 308.2}{8.04} \approx \frac{80 \times 300}{8} \approx 3000$

Note: The last digit in 4-figure tables is frequently not accurate. It is usual, therefore, to round final answers to 2 significant figures or 3 significant figures.

Exercise 1c (Revision)

Use tables to calculate the following. Give answers to 3 s.f.

- | | |
|------------------------------------|--|
| ① 29.3×142 | ② $31.6 \div 4.79$ |
| ③ $4627 \div 81.5$ | ④ 193×279 |
| ⑤ 45.6×40.9 | ⑥ $818.3 \div 2.905$ |
| ⑦ $\frac{45.6 \times 40.9}{72.5}$ | ⑧ $\frac{818.3 \times 72.5}{2.905}$ |
| ⑨ $\frac{67800}{8.71 \times 36.8}$ | ⑩ $\frac{7820 \times 5.05}{944 \times 11.7}$ |

Powers and roots (revision)

To calculate powers and roots of numbers, first express them as logarithms and then apply the multiplication and division laws of indices to the logarithms. As before, it is most convenient if the work is set out in table form.

Example 3

Evaluate **a** 3.55^4 , **b** $\sqrt[5]{40\,000}$.

number	log
3.55^4	$\rightarrow 0.5502 \times 4$
158.8	$\leftarrow 2.2008$

$$3.55^4 = 158.8$$

number	log
$\sqrt[5]{40\,000}$	$\rightarrow 4.6021 \div 5$
8.326	$\leftarrow 0.9204$

$$\sqrt[5]{40\,000} = 8.326$$

Example 4

Evaluate $\frac{\sqrt{94\,100} \times 38.2}{5.68^3 \times 8.14}$ correct to 2 significant figures.

Method

Find a single logarithm representing the numerator and a single logarithm representing the denominator. Subtract log-denominator from log-numerator.

number	log	
$\sqrt{94\,100}$	$\rightarrow 4.9736 \div 2 = 2.4868$	
38.2	$\rightarrow$	+1.5821
numerator		4.0689
		4.0689
5.68^3	$\rightarrow 0.7543 \times 3 = 2.2629$	
8.14	$\rightarrow$	+0.9106
denominator		3.1735
		-3.1735
7.859	$\leftarrow$	0.8954

$$\frac{\sqrt{94\,100} \times 38.2}{5.68^3 \times 8.14} = 7.859$$

$$= 7.9 \text{ to 2 s.f.}$$

Exercise 1d (Revision)

Evaluate the following, rounding off each answer to 3 significant figures.

① 9.21^4

② 82.5^2

③ 572^3

④ $\sqrt{62.1}$

⑤ $\sqrt[5]{93.6}$

⑥ $\sqrt[3]{1\,630}$

⑦ $\sqrt[4]{85.3 \times 7.28}$

⑧ $\left(\frac{184.8}{6.2}\right)^3$

⑨ $\frac{27.1^2 \times 327}{\sqrt{27\,500\,000}}$

⑩ $\frac{95.3 \times \sqrt[3]{18.4}}{1.29^5 \times 2.03}$

1.3 Logarithms of numbers less than 1

To find the logarithms of numbers less than 1 we use negative powers of 10. For example,

$$\begin{aligned} 0.037 &= 3.7 \times 0.01 \\ &= 10^{0.5682} \times 10^{-2} \\ &= 10^{0.5682 + (-2)} \\ &= 10^{-2 + 0.5682} \end{aligned}$$

$$\log 0.037 = -2 + 0.5682$$

In practice, the logarithm of 0.037 is written as $\bar{2}.5682$ and said as 'bar 2 point 5682'.

Notice that $\bar{2}.5682$ is *not* the same as -2.5682 :

$$\begin{aligned} \bar{2}.5682 &\text{ means } -2 + 0.5682, \\ -2.5682 &\text{ means } -2 + (-0.5682). \end{aligned}$$

The logarithm $\bar{2}.5682$ has a negative characteristic ($\bar{2}$) and a positive mantissa (0.5682). As before, we find the characteristic by expressing the given number in standard form. For example,

$$\begin{aligned} \log 0.04132 &= \log (4.132 \times 10^{-2}) \\ &\text{the characteristic is } \bar{2} \end{aligned}$$

$$\begin{aligned} \log 0.00087 &= \log (8.7 \times 10^{-4}) \\ &\text{the characteristic is } \bar{4}. \end{aligned}$$

Example 5

a Express 0.005812 as a number in standard form.

b Hence use tables to find the logarithm of 0.005812.

$$\begin{aligned} \text{a } 0.005812 &= 5.812 \times 0.001 \\ &= 5.812 \times 10^{-3} \end{aligned}$$

$$\begin{aligned} \text{b } \log 0.005812 &= \log (5.812 \times 10^{-3}) \\ &= \log 5.813 + (-3) \\ &= 0.7643 + (-3) \\ &\quad \text{(from log tables)} \\ &= -3 + 0.7643 \\ &= \bar{3}.7643 \end{aligned}$$